BANDA SRI HARI

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GitHub | LinkedIn | Portfolio

EDUCATION

Sir M. Visvesvaraya Institute of Technology, Bengaluru — **Expected 2027**
B.E. in Artificial Intelligence & Machine Learning — **CGPA: 8.72/10**

EXPERIENCE

Chiacon Consulting Private Limited — **Mar 2026 – Jun 2026**
AI Engineer Intern — AI Voice Interview & HR Orchestration Platform — Remote, India

- Engineered a real-time Python/FastAPI WebSocket voice-interview backend integrating Recall.ai and Azure Speech with thread-safe SDK callbacks and continuous audio streaming.
- Architected a 9-node LangGraph workflow with typed state, deterministic routing, project deduplication, guardrails, and bounded memory for multi-turn interview control.
- Built concurrent Technical, Behavioral, and Integrity LLM evaluators with `asyncio.gather()`, enforcing Pydantic/OpenAI Function Calling structured outputs.
- Integrated Azure OpenAI, Microsoft Graph, Entra ID, Recall.ai, and Azure Communication Services via REST APIs with OAuth 2.0/HMAC; added a raw REST workaround for an SDK limitation.
- Developed a headless Python regression harness with 7 synthetic personas, LLM simulation, `unittest.mock`, and 5 assertions; debugged state-machine, SDK, acoustic-feedback, and file-write failures.

SKILLS

Languages: Python, SQL, JavaScript
Backend & APIs: FastAPI, REST APIs, WebSockets, Webhooks, OAuth 2.0, Pydantic, Async Programming
AI / GenAI Engineering: LLMs, Generative AI, RAG, Embeddings, Vector Search, AI Agents, Conversational AI, LangGraph, LangChain, MCP, LLM-as-a-Judge Evaluation, Structured Outputs, Function Calling, Context Engineering, Guardrails
Databases & Retrieval: PostgreSQL, pgvector, FAISS, SQLite, DynamoDB, Hybrid Retrieval, Tree-sitter/AST
Cloud & Deployment: AWS Lambda, API Gateway, S3, Amazon Bedrock, Bedrock Knowledge Bases, Docker, GitHub Actions
Testing & Reliability: Pytest, Regression Testing, Mocking, Debugging

PROJECTS

GitNova – Open-Source Contribution Intelligence Platform — Live | GitHub

- Built a full-stack code-intelligence platform with Python/FastAPI, PostgreSQL/pgvector, React, GitHub REST ingestion, Docker, and scheduled GitHub Actions RAG evaluation.
- Engineered hybrid code RAG over 41,000+ AST-indexed chunks using code embeddings, pgvector, PostgreSQL full-text search, Tree-sitter/AST parsing, and Reciprocal Rank Fusion.
- Designed a dual-phase Gemini/Groq reasoning pipeline with Pydantic outputs and fail-closed `GroundingVerifier` checks for files, symbols, test commands, and repository evidence before publication.
- Built automated tests across API, database, AST/retrieval, LLM resilience, and publication guardrails; resolved pgBouncer, PostgREST RPC, and cloud build-memory failures.

Stack: Python, FastAPI, PostgreSQL, pgvector, React, Tree-sitter, Gemini, Groq, GitHub Actions, Docker

Orbit – Serverless AI Learning Platform — GitHub

- Architected a serverless AI learning platform with Python/FastAPI on AWS Lambda/Mangum, exposing 8 API Gateway endpoints for code evaluation, viva, session, curriculum, and mentor workflows.
- Engineered DynamoDB assessment state with composite keys, atomic updates, and state-machine transitions for attempts, viva locking, and learning progression without client-side state.
- Built managed RAG with S3 and Bedrock Knowledge Bases, ingesting timestamped YouTube lecture transcripts for semantic retrieval and grounded mentor responses.
- Integrated Bedrock and Guardrails for code evaluation, viva generation/grading, and mentor interactions; resolved API Gateway timeouts, Guardrail false positives, and non-JSON errors with defensive fallbacks.

Stack: Python, FastAPI, AWS Lambda, Mangum, API Gateway, DynamoDB, S3, Bedrock

Unshell – AML & KYB Forensic Intelligence Graph — *Hackfest 2026 – 2nd Runner-Up* | GitHub

- Built an AML/KYB forensic intelligence system using Python/FastAPI, Companies House REST data, PDF RAG, deterministic NetworkX analysis, and OFAC SDN screening to trace hidden ownership.
- Engineered a deterministic NetworkX risk engine using Johnson's cycle detection, degree/path-depth checks, and 11 rule-based typologies for circular ownership, nominee pass-throughs, and layered structures.
- Designed a fail-closed grounding pipeline with PyMuPDF, `all-MiniLM-L6-v2`, FAISS, Gemini structured outputs, and RapidFuzz to validate entities and evidence before graph analysis.
- Built resilient multi-source ingestion with recursive Level-2 Companies House traversal, rate-limit handling, and indexed SQLite/`aiosqlite` OFAC SDN screening.

Stack: Python, FastAPI, LangGraph, PyMuPDF, FAISS, Sentence-Transformers, Gemini, RapidFuzz, NetworkX, SQLite, FastMCP

ACHIEVEMENTS & CERTIFICATIONS

- **Hackfest 2026:** 2nd Runner-Up – Top 50 / 750+ teams
- **Amazon ML Challenge 2025:** Top 15% among 10,000+ teams
- **NPTEL:** Mathematical Foundations for Machine Learning – Elite Silver (IIT Guwahati)
- **Oracle Certified:** Generative AI Professional, Data Science Professional, Database@AWS Foundations Associate